

◆ 16.11.4 PAC-Based Decision Integration (Final — Integrated Version)

Triadic Representation

$$X = (R, C, P)$$

where:

- R: measurement (reality)
- C: computation (inference)
- P: prediction (model)

Independence and Boundedness

Assume weak dependence:

$$|\mathrm{Corr}(R, C)|, |\mathrm{Corr}(R, P)|, |\mathrm{Corr}(C, P)| \leq \delta, \quad \delta \ll 1$$

and bounded signals:

$$R, C, P \in [m, M]$$

Thus:

the triadic state is compact and free from redundancy amplification.

Robust Fusion Mechanism

A robust aggregation is defined as:

$$D = \mathrm{median}(R, C, P)$$

Thus:

the fusion rule achieves robustness with breakdown point up to 33\%.

PAC Confidence Measure

$$\Gamma = KL(P \parallel P_0)$$

$$PAC = \frac{1}{1 + e^{-\Gamma}}$$

Thus:

$$PAC \in (0,1)$$

and evolves smoothly:

$$|PAC_{k+1} - PAC_k| \leq L \Delta t$$

Adaptive Weighted Fusion

$D =$

$\sum_{i \in \{R,C,P\}} w_i X_i,$

$\quad$

$w_i =$

$\frac{PAC_i}{\sum_j PAC_j + \epsilon}$

$\sum_i w_i = 1$

Thus:

higher-confidence components contribute more to the decision.

Integration into Dynamics

$F \supset \alpha(PAC) \nabla \ln P,$

$\quad$

$0 < \alpha(PAC) \leq \alpha_{\max}$

Thus:

probabilistic confidence directly modulates system evolution.

Network-Level Coupling

$$w_{ij} \propto PAC_i \cdot PAC_j$$

Thus:

reliable agents dominate the network influence.

Stability Compatibility

Since:

- PAC is bounded
- weights w_i are normalized and bounded
- fusion D is bounded

Thus:

the PAC mechanism preserves Lyapunov stability and does not introduce destabilizing effects.

Implication

PAC provides a bounded, robust, and bias-

resistant decision fusion mechanism, fully compatible with the closed-loop dynamics and stability guarantees of the SEKHEM-TSDD framework.

**SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
DOI: 10.5281/zenodo.19521219"**